DocuMind AI

Intelligent Document Management System — Explained Simply

Friendly Companion Guide to the Technical README

Welcome, team! This document explains our Capstone Project in plain English. No coding experience required!

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1. What is DocuMind AI?

In one sentence: DocuMind AI is a smart document storage platform that uses artificial intelligence to automatically read your files and answer your questions about them instantly.

Imagine having a huge stack of PDF reports or Word documents. Instead of reading every single page to find what you need, you upload them to DocuMind. The system reads

them for you, creates summaries, and lets you "chat" with the documents to get instant answers.

> **Think of it like...**
>
> *A magical librarian. You hand the librarian a 50-page book. Within seconds, the librarian memorizes the entire book, hands you a short summary, and says, "Ask me anything about what's inside!"*

2. Why This Project Exists

We built this system to solve a very common problem: **information overload**. People deal with long business reports, research papers, and legal documents every day. Manually searching through them takes hours. DocuMind AI exists to save time, help organize information beautifully, and make sharing knowledge with teammates incredibly easy.

3. What the System Can Do

Upload & Store Securely

Users can safely upload PDF and DOCX (Word) files. The system keeps them secure and organized.

Automatic Summaries & Keywords

The AI automatically writes a 2-3 paragraph summary and picks out the 5-10 most important topics.

Chat With Documents

Type a question like, "What is our total revenue?" and the AI will search the document and give you the answer.

Collections & Organization

Create color-coded folders called "Collections" to group similar documents together.

Share & Collaborate

Securely share files with specific teammates, let them view the documents, and track who accessed what.

Advanced Search

Instantly search across every approved document in the system using lightning-fast full-text search.

4. A Simple User Story

Let's follow a new user named Maya to see how the platform works:

1. **Registration:** Maya creates an account and logs into her fresh dashboard.
2. **Upload:** She uploads a 50-page Quarterly Business Report.
3. **Waiting:** The document goes into "Pending" status. An Admin logs in, checks to make sure the file is safe, and clicks "Approve."
4. **AI Magic:** Instantly, the AI reads Maya's report. It generates a summary and pulls out keywords like *Revenue* and *Expenses*.
5. **Chatting:** Maya asks the AI, "Did we meet our sales targets?" The AI replies with a detailed answer from page 14.
6. **Sharing:** Maya shares the document with her manager, who can now log in and ask their own questions!

5. Regular Users vs. Admins

DocuMind has two main types of users. Here is a simple comparison of what they can do:

Feature / Action	Regular User	Admin User
Upload Documents	Yes (Up to 10 files by default)	Yes
Ask the AI Questions	Yes (Up to 20 questions a day)	Yes

Feature / Action	Regular User	Admin User
Approve & Reject Documents	No	Yes (They are the gatekeepers)
View All System Documents	No (Only their own & shared)	Yes
Change User Limits (Quotas)	No	Yes (Can give power users more uploads)
View Analytics Dashboard	No	Yes (Can see top contributors and activity)

6. Behind the Scenes (How it Works)

When you click "Upload," here is the hidden journey your file takes:

1. **Validation:** The system checks: *Is this a PDF or Word doc? Is it under 20MB?*
2. **Text Extraction:** The system cracks open the file and pulls out all the readable text.
3. **Database Save:** The text is saved safely into our giant digital filing cabinet (MySQL).
4. **Pending Approval:** The document waits patiently in a holding area.
5. **Admin Review:** An Admin gives the thumbs up.
6. **AI Analysis:** The AI reads the text, creates the summary, and generates keywords automatically.
7. **Ready to Use!** The document is now active for chatting and sharing.

7. The AI Explained Simply

We use a highly advanced AI created by Google called **Gemini**. It is incredibly smart, but it has specific rules and limits.

- **Response Styles:** You can tell the AI how you want it to answer: *Concise* (short bullets), *Balanced* (clear paragraphs), or *Detailed* (in-depth explanations).
- **Context Awareness:** The AI remembers what you just asked! If you ask, "Who is the CEO?", and then ask, "How old is he?", the AI knows you are still talking about the CEO.
- **Summarization Limit:** For the automatic summary, the AI only reads the first 12,000 characters (the first few pages).

- **Document Q&A Limit:** When you ask a question, it searches the main chunks of the text, but extremely massive documents might lose some details at the very end.
- **No Pictures:** The AI only reads text. It cannot look at charts, graphs, or scanned images (No OCR).
- **Language:** It works best in English.

> **Important Warning (Hallucinations):**
>
> *Sometimes, AI tries so hard to be helpful that it accidentally makes up an answer if it can't find one. This is called a "hallucination." Always double-check important facts (like legal rules or budget numbers) by looking at the original document!*

8. Limits and Quotas

To keep the system running fast and avoid huge AI bills, we put some fair limits in place (which Admins can change):

Limit Type	The Rule	Why?
File Size	20 MB per file maximum.	Keeps our servers from getting clogged.
Upload Limit	10 documents per regular user.	Saves storage space. (If you delete an old file, you get a spot back!)
AI Questions	20 questions per day.	Google charges for AI usage. The limit resets every single day.

9. Security Made Simple

Keeping documents safe is our top priority. Here is how we protect the system:

> **Think of it like home security...**
>
> - **Passwords (Encryption):**
> *We scramble passwords into a random mess (like a secret riddle). Even our Admins cannot see your real password.*
> - **SQL Injection Prevention:**

Hackers try to break databases by typing computer code into login boxes. We use special forms that act like a mail slot—they only accept standard letters, not crowbars!

- **CSRF Protection:**

  Every form has a hidden secret stamp. If a fake website tries to send a form to us, it won't have the stamp, so we throw it away.

- **XSS Protection:**

  We "wash" all text that users type. If someone types a virus code into a comment, we wash it so it just displays as harmless text.

- **API Key Safety:**

  Our secret password to use Google's AI is locked inside a hidden vault (a `.env` file). Nobody on the internet can see it.

Furthermore, we use **Sessions** (which help remember who you are and can time out after inactivity) and **Protected Routes** (you cannot visit private dashboard pages without logging in first). The README also makes an important distinction: the **Public Library** can be viewed without logging in, but most personal and admin pages are locked behind authentication.

10. Our Technical Stack

These are the tools we used to build the house:

- **PHP:** The main brain/engine of our application on the server.
- **MySQL & PDO:** The digital filing cabinet where all data is stored securely.
- **HTML, CSS & JavaScript:** The paint, layout, and interactive buttons that make the website look great in your browser.
- **Bootstrap/Tailwind:** Pre-made design styles to make things look professional quickly.
- **Composer:** A helpful tool that fetches extra pieces of code we need from the internet.
- **Google Gemini API:** The phone line we use to talk to Google's AI.

11. Project Structure Map

If DocuMind AI was a large school, here is what the rooms would be:

Think of it like a school layout...

- **app/Controllers:**

  The Principals. They take your requests and tell the rest of the school what to do.

- **app/Models:**

  The Filing Clerks. They are the only ones allowed to talk to the MySQL Database.

- **app/Services:**

  The Specialists. This room holds our AI expert and the Document Parser (who extracts text).

- **app/Middleware:**

  The Security Guards. They check your ID badge before letting you into certain rooms.

- **app/Helpers:**

  The Janitors. They do small, helpful utility tasks around the school.

- **config/ & .env:**

  The Master Rulebook and Key Safe. Holds all our private settings.

- **database/:**

  The Blueprints. Instructions on how to build our filing cabinets.

- **public/:**

  The Front Lobby. The only place outside visitors are allowed to step into.

- **storage/:**

  The Basement Vault. Where the actual uploaded PDF files live safely.

- **views/:**

  The Decorated Classrooms. The HTML files that determine what you actually see on your screen.

- **vendor/:**

  The Supply Closet. Where Composer puts all the borrowed tools.

12. Simple Setup Overview

If a person needs to run this on their own computer, here is the quick checklist:

1. Copy the project folder into your local web server (like XAMPP).
2. Run a command called `composer install` to fill up the supply closet.
3. Copy the `.env.example` file, rename it to `.env`, and add your database passwords and Google API key.

4. Create an empty database named `documind` and import our blueprint (`schema.sql`).
5. Make sure the `storage` folder is writable, then start your server!

13. Troubleshooting Cheatsheet

- **Database Connection Error?** You probably typed the wrong password or database name in your `.env` file.
- **"Composer packages not found"?** You forgot to run `composer install`.
- **AI features failing?** Check that your Google Gemini API key is correct and active.
- **Can't upload a file?** Make sure it is exactly a PDF or DOCX, and under 20MB! Also check that your storage folder allows writing.
- **White Screen of Death (500 Error)?** Check the `storage/logs/` folder. The application wrote down exactly what broke there!

14. Limitations & Future Updates

We are super proud of DocuMind AI, but there is always room to grow! Currently, the system **cannot** read scanned pictures (No OCR), there is no mobile app, and users cannot edit a document together at the exact same time.

In the future, we plan to add:

- A Dark Mode interface.
- Advanced OCR so the AI can read text hidden inside images.
- Real-time chat so you and a teammate can talk to the AI at the same time.
- Multi-language AI responses!

15. The "Top 5" Recap

If you remember only 5 things from this whole guide, remember these:

1. **We Read For You:** DocuMind AI extracts text from PDFs and Word docs automatically.
2. **Admins Rule:** All uploaded files must be approved by an Admin before the AI touches them.
3. **Instant Answers:** The AI summarizes documents and answers your questions using Google Gemini.
4. **Daily Limits:** Regular users normally have a 10-document upload limit and 20 AI questions per day to save resources.

5. **Highly Secure:** We use top-tier security to protect passwords, block hackers, and hide our API keys.

16. Tiny Glossary

- **AI (Artificial Intelligence):** A computer brain trained to read text and answer questions like a human.
- **API:** A digital bridge that lets our website talk to Google's computers.
- **Database:** The highly organized digital filing cabinet where user data is saved.
- **Collection:** A virtual folder you create to organize related documents.
- **Middleware:** Code that acts as a security checkpoint before loading a webpage.
- **Admin:** A boss-level user who manages the system and approves documents.